



TITAN GRID: A SMART FACTORY FOR NEAR-SHORING THE PRODUCTION OF TRANSFORMER TANKS AND RADIATOR FIN WALLS



FIN-2.4 OPEX & Labor Matrix

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FIN-2.4: Operating Cost Structure & Labor Matrix

TITAN Grid – Cost Engine & Margin Stability Framework

Source Document:

Projected Annual OPEX (Steady State): €22.5M – €24.0M

Primary Cost Driver: Raw Materials (Steel) ~70% of OPEX

Target EBITDA Margin: 28% – 32%

Executive Summary: A Capital-Intensive, Not Labor-Intensive Model

TITAN Grid is designed as a technology-driven industrial platform, where:

- **Costs are variable and controllable**
- **Labor exposure is structurally minimized**
- **Margins are protected through pricing mechanisms**

The cost structure is built for resilience—profitability is sustained even amidst volatility.

1. Labor Matrix: High Productivity, Low Cost Exposure

In contrast to conventional manufacturing, TITAN Grid generates €34M+ in revenue with just ~50 employees, thanks to automation.

Workforce Structure

Category	Headcount	Avg. Net Salary (€)	Annual Gross (€)	Strategic Role
Top Management	3	3,500	185,000	Strategy, P&L, OEM relations
Engineers (IT / Robotics / QA)	7	2,200	275,000	Digital Twin, MES, certification
CNC & Laser Operators	10	1,400	250,000	Precision cutting
Robotic Welding Supervisors	15	1,500	400,000	Automated welding oversight
Logistics & Maintenance	10	1,000	180,000	Supply chain & equipment uptime
Admin & Sales	5	1,200	110,000	Back-office & reporting





**TITAN GRID – ADVANCED
TRANSFORMER TANK
MANUFACTURING FACILITY**

TRANSFORMER TANKS & FIN WALLS

Metal D.O.O.

Total Workforce

- Approximately 50 employees
- Total Labor Cost: Around €1.4 million annually

Key Insight

- Labor accounts for approximately 4–5% of total revenue

→ This is exceptionally low when compared to traditional manufacturing, which typically ranges from 15–25%.

2. Non-Labor OPEX Structure (Cost Drivers)

The cost structure is primarily driven by pass-through variable costs, which help maintain margin stability.

Cost Breakdown

Cost Category	Annual (€)	Strategic Role
Raw Materials (Steel)	18,500,000	Primary input (pass-through protected)
Energy (Electricity & Gas)	850,000	Reduced via solar (-20%)
Maintenance (Industry 4.0)	450,000	Software + machine servicing
Insurance & Security	120,000	Asset and logistics protection
Marketing & R&D	250,000	Pipeline expansion (TITAN 2 & 3)
Administrative Costs	150,000	Overhead (audit, telecom, travel)

Key Insight

- Steel dominates cost structure (~70%)
- Fully aligned with pass-through pricing model

→ Protects EBITDA margins from volatility





3. Cost Structure Logic: Fixed vs. Variable

Cost Composition

Type	Share	Interpretation
Variable Costs	~85%	Flexible, volume-linked
Fixed Costs	~15%	Low operational rigidity

Strategic Implication

- Enables:
 - Low break-even point (24–27%)
 - Rapid cost adjustment during downturns
- Core driver of “Safety First” model

4. Efficiency Ratios (Investor KPIs)

Key Performance Indicators

Metric	Value	Interpretation
Revenue per Employee	€690K – €760K	Top-tier European benchmark
Labor Cost / Revenue	~4%	Extremely efficient
EBITDA Margin	28–32%	High-margin industrial profile

Investor Insight

→ This is not a labor business—it is a capital + technology business





5. Margin Stability Mechanism

Why EBITDA Remains Stable (~30%)

1. Steel Pass-Through

- Raw material costs are passed on to the customer.

2. Low Labor Exposure

- Wage inflation has minimal impact.

3. Energy Hedge

- Solar energy helps to mitigate cost volatility.

4. Automation Efficiency

- Ensures consistent production quality.

Result

→ EBITDA stays stable throughout market cycles.

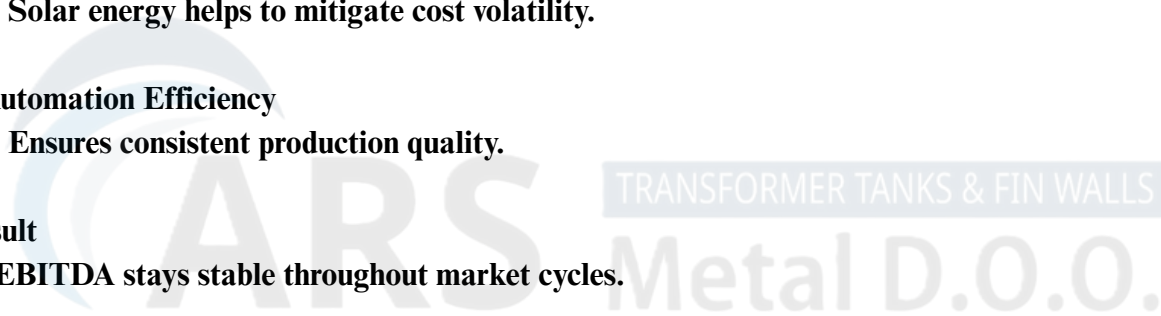
6. Connection to Credit Strength (Impact on DSCR)

The OPEX structure directly enhances:

- DSCR: 1.58x
- Stable CFADS (Cash Flow Available for Debt Service)

Why Lenders Favor This Model

- Predictable cost structure
- Limited risk regarding labor disruptions
- Strong margin protection





7. Risk Mitigation Incorporated in OPEX

Risk	Mitigation
Labor inflation	Low labor share (~4%)
Steel volatility	Pass-through pricing
Energy price spikes	Solar hedge
Downtime risk	Predictive maintenance

8. Strategic Conclusion: Cost as a Competitive Advantage

“TITAN Grid’s cost structure converts traditional industrial risks into manageable variables—guaranteeing stable margins, a low break-even point, and predictable cash flow.”

Final Investment Insight

This OPEX model provides:

- **High profitability (EBITDA ~30%)**
- **Low operational risk**
- **Strong resilience in challenging scenarios**

→ Directly supporting:

- **IRR: 32.8%**
- **DSCR: 1.58x**
- **Break-even:** <27% capacity**

